

Structured Local Adaptation and Hybrid Adjudication in Ultra-Low-Resource Tangut Short-Text Translation

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Abstract

Tangut translation is an extreme low-resource problem: the current repository contains only 491 real Tangut-Chinese pairs, split 400/41/50 into train/dev/test, and many targets are compact title-like Chinese strings rather than ordinary sentences. We compare local prompt-only baselines, a stronger frontier prompt-only baseline, synthetic-data supervised fine-tuning (SFT), direct preference optimization (DPO), and inference-time hybrid adjudication. DeepSeek-V3.2 with dictionary grounding and four fixed few-shot exemplars is the strongest single model, reaching 2.80/5 on a reference-aware judge with 12/50 exact matches. Within the open/local regime, structured multitask SFT is the best supervised base, and DPO helps only on that base with strict preference-gap filtering: the gap-0.4 sigmoid variant reaches 30.01 chrF++ and is the strongest local model. A catalog-aware adjudicator over the frontier output, UNK-SFT, and that DPO model yields the strongest non-reference system overall, with 34.87 chrF++, 3.02 reference-aware overall, 14/50 exact matches, and zero contamination. Bootstrap resampling confirms a robust chrF++ gain over the frontier baseline (+6.33, 95% CI [2.10, 11.67]); judge and exact-match gains are positive but boundary-level on this 50-item test set. A local Qwen judge preserves the same frontier < hybrid < human ordering, and a deterministic open selector over the same pool still reaches 32.62 chrF++, showing that part of the workflow gain is reproducible without closed adjudication. Finally, lexical coverage and perplexity mis-rank the gold reference itself, so reference-aware evidence should be treated as primary and reference-free metrics as diagnostic. The practical lesson is two-level: strong frontier prompting is the best single-model upper bound, but structured local training remains useful because it supplies complementary historical-title hypotheses for a stronger scholar-facing workflow.

1 Introduction

Machine translation for historical ultra-low-resource languages is hard for two separate reasons: parallel data are scarce, and the outputs that matter to scholars are often short titles or gloss-like strings rather than fluent modern sentences. Recent work on ancient-language NLP and historical-text translation has made this setting more visible, but it has also shown that both supervision and evaluation become unstable outside modern benchmark conditions [1, 2, 3]. Tangut is a canonical example. In the current repository snapshot, only 491 real Tangut-Chinese pairs are available.

This regime also creates an obvious objection: perhaps a strong Chinese frontier LLM, given dictionary glosses and a careful title-oriented prompt, is already enough. We therefore compare five components in one frame: local dictionary-grounded prompting, a stronger proprietary frontier prompt-only baseline, synthetic pseudo-parallel SFT, preference optimization on top of SFT, and a new inference-time hybrid adjudication workflow [5, 4, 11]. The point is to separate what prompting already buys, what local training still buys, and whether the two can be combined into a stronger scholar-facing system.

Our findings are straightforward. The stronger prompt-only objection is real: the DeepSeek-V3.2 few-shot dictionary baseline is the strongest single model we test, with 2.80/5 overall under the

reference-aware judge and 12/50 exact matches. Within the open/local trainable regime, structured multitask SFT is the right base, and DPO helps only after it is moved onto that base and the preference pairs are filtered much more aggressively; the strict gap-filtered sigmoid variant reaches the highest local chrF++ in the repository (30.01). Complementarity is also not merely an oracle observation. A catalog-aware three-way adjudicator over the frontier output, UNK-SFT, and the strict gap-filtered DPO model reaches 34.87 chrF++, 3.02 overall, and 14 exact matches, while a deterministic open selector over the same pool still recovers part of that gain without any closed adjudicator. Finally, the repository exposes two robustness issues directly: several reference-free metrics mis-rank the gold reference itself, and low-gap preference data remain noisy enough to destabilize DPO.

The current draft therefore makes three contributions:

1. an empirical comparison that cleanly separates the strongest single model, the strongest local trainable recipe, and the strongest hybrid workflow under only 491 real pairs;
2. a positive local-method result showing that structured multitask SFT is the right base and that strict gap-filtered DPO produces the strongest local model, while catalog-aware hybrid adjudication turns those local candidates into a better workflow and a deterministic open selector recovers part of the same gain;
3. a reference-aware evaluation, robustness, and data-hygiene analysis showing why reference-free metrics and low-gap preference data can mislead in this task.

The rest of the paper situates the task against related work, describes the comparison families, and then keeps two claims separate: how to build the best open Tangut model, and how to combine frontier and local candidates into a stronger workflow.

2 Related Work

Ancient and historical language processing. Ancient-language NLP differs from mainstream NLP not only because data are scarce, but also because scripts, editorial conventions, and annotation practices are unstable across corpora [1]. Shared tasks on historical translation and document modernization report similar sparsity, though they often target broader sentence translation or orthographic updating rather than cross-script title reconstruction [2, 3]. Our Tangut setting is narrower still: only 491 real pairs are available, and the evaluation set is dominated by compact title-like strings.

Low-resource MT, synthetic supervision, and lexical grounding. Low-resource MT often relies on synthetic parallel data, multilingual transfer, or explicit lexical constraints [4, 6, 8, 5, 7]. Tangut overlaps with this literature but is harsher in two ways: the source script is largely outside standard MT pretraining, and the outputs are compact catalog titles rather than ordinary sentences. We therefore focus on synthetic supervision and lexicon-grounded structure rather than on broad multilingual transfer.

Dictionary-structured prompting and error-aware prompting. Recent LLM translation work suggests that lexical resources are most useful when presented as structured intermediate evidence rather than as flat prompt context. Chain-of-Dictionary prompting and Learning-From-Mistakes prompting both follow this pattern in low-resource settings [9, 10]. Our stronger frontier baseline is adjacent to this line, but the present paper asks a different question: whether alignment

structure should be absorbed into trainable local targets, and whether those local hypotheses remain useful after a strong frontier prompt-only baseline is already in place.

Preference optimization. Direct Preference Optimization (DPO) is now a standard offline alignment objective, and recent work has extended it to machine translation and noisy-preference settings [11, 14, 15, 12, 13]. Our results place this literature in a harsher regime: supervision is tiny, outputs are title-like, and the chosen/rejected signal is built from partially misaligned automatic metrics. In this setting, DPO helps only after base-model choice and pair filtering are controlled aggressively.

Evaluation for non-standard translation outputs. chrF++ is widely used in machine translation because character-level overlap is often more stable than token-level overlap under orthographic variation [16]. Learned MT metrics and QE systems such as COMET and OpenKiwi go beyond surface overlap, but recent work also shows that they can have domain-specific failure modes and induce metric interference [17, 20, 18, 19]. Our findings sharpen that caution in a historical short-text regime: lexical coverage and perplexity can penalize the gold reference itself, so reference-aware evidence must remain primary.

3 Task Setting

The real Tangut data in this repository contains 491 pairs. The current split is 400 training examples, 41 development examples, and 50 test examples, produced by a single random shuffle with seed 42. Unlike ordinary machine-translation benchmarks, the targets are short and often title-like. Many examples resemble book titles, sutra titles, or compact scholastic expressions. This matters because title reconstruction rewards precise lexical choices and compact formatting, while a modern Chinese fluency model may incorrectly treat those outputs as unnatural.

Because the real dataset is tiny, the system uses synthetic data to train translation models. For each SFT branch, 50,000 synthetic Tangut-Chinese pairs are produced from monolingual classical Chinese and dictionary knowledge, then combined with the 400 real training examples upsampled by a factor of 10 to form a 54,000-example training set. All learned models use the same instruction-tuned 7B backbone. These synthetic inputs are pseudo-Tangut training proxies rather than claims about faithful Tangut reconstruction; in other words, we judge them only by downstream utility on held-out real Tangut titles, not by intrinsic linguistic fidelity.

We also explicitly audit data hygiene. There is zero exact source, target, or full-pair overlap across the train/dev/test splits, zero exact overlap between the 50 test references and the final modern-Chinese targets of the 50k synthetic multitask corpus, and zero exact overlap between the four development examples used in the frontier few-shot prompt and the test references. Character n -gram overlap is naturally non-zero inside the real dataset because titles share common Buddhist morphemes, but the synthetic multitask targets are far from copied test references: their mean maximum Jaccard overlap with test references is 0.027 for character bigrams, 0.004 for trigrams, and 0.000 for 4-grams.

We evaluate with two classes of metrics. The repository’s existing metrics are lexical coverage, perplexity, chrF++, and a dictionary-conditioned LLM judge. chrF++ is a reasonable overlap anchor for this setting because character-based scoring is relatively stable under orthographic variation [16]. Perplexity is computed as $\exp(\ell)$, where ℓ is the mean token-level causal-LM loss under Qwen2.5-0.5B. In this title-heavy setting that score becomes extremely large even for the gold reference, so we treat it as diagnostic rather than dispositive. “Contamination” in the current

paper likewise needs a narrow definition: it means mixed-script or artifact contamination in the *output* string, such as Latin letters, literal [UNK] tokens, or prompt residue, not train-test data leakage. Learned MT evaluation and quality-estimation systems can capture richer signals than surface overlap [17, 20], but recent work also shows that strong neural MT metrics can have hidden domain-specific failure modes and can induce metric interference when similar signals appear in both system development and final evaluation [18, 19]. For that reason, we do not use off-the-shelf COMET as a headline arbiter in this draft: standard checkpoints are designed for modern sentence-level MT rather than Tangut-script short-title reconstruction, so any COMET score would still require task-specific validation. Accordingly, this draft does not treat all metrics equally: chrF++, exact match, contamination, suffix preservation, and the new reference-aware judge are primary evidence, while lexical coverage, perplexity, and the legacy reference-free judge are reported as diagnostics.

4 Methods

4.1 Inference-only baselines

The first family contains no weight updates. **baseline1** prompts the local model directly on the Tangut string; **baseline2** adds dictionary glosses; and **baseline2_1_cot** further decomposes generation into alignment, literal drafting, and rewriting. This family is closest to lexicon-augmented MT, except that the lexicon is injected through prompting rather than decoder constraints [5, 7].

We also add a deliberately stronger prompt-only comparison based on DeepSeek-V3.2 with dictionary grounding and four fixed development-set exemplars (IDs 1, 6, 15, and 16: 佛說大人八覺經, 佛說菩薩行修經, 金光明總持經, and 摩訶般若波羅蜜多心經). These cover a short canonical sutra title, a bodhisattva-practice title, a dharani-style title, and a long transliteration-heavy chain. The prompt emphasizes compact title form, forbids explanatory paraphrase, and emits only the final title string. Decoding uses temperature 0.0 with a 512-token cap. The goal is not to build a straw baseline but to test directly whether frontier prompting already solves the task well enough.

4.2 Synthetic SFT baselines

The second family uses synthetic training data. We build 50k pseudo-Tangut/Chinese pairs from monolingual historical Chinese by mapping Chinese characters to Tangut characters through dictionary explanations, fixing the synthesis seed to 42, and keeping source lengths between 4 and 64 characters. The four branches differ only in how dictionary misses are handled and whether the target includes alignment structure: mixed-script keeps 30–70% Chinese characters, UNK replaces misses with [UNK], semantic projection replaces misses through Qwen2.5-0.5B embeddings plus FAISS lookup, and multitask keeps the UNK input but changes the target to `<dict_match>{...}</dict_match><literal>{...}</literal> {final title}`. This gives the model both an alignment channel and a final title target in one sequence. These examples are not philologically faithful Tangut reconstructions; they are structured training proxies. Each synthetic set is combined with the 400 real training examples upsampled by a factor of 10 and shuffled with seed 42. The key design question is whether the model benefits more from input purity, approximate semantic preservation, or explicit alignment structure.

4.3 Preference optimization

The final family applies DPO to an SFT base using reward-defined chosen/rejected pairs [11]. We sample five outputs per training prompt at temperature 0.8 and top- p 0.95, then score them with

$$r(y) = \text{lex}(y) - 0.01 \log \text{ppl}(y),$$

where $\text{lex}(y)$ is the repository’s lexical-coverage score. Concretely, lexical coverage segments the Chinese output with jieba plus single-character backoff, performs maximum forward matching over the Tangut input with dictionary entries up to length five, and reports the fraction of matched Tangut spans whose gloss set intersects the output token set. The $\log \text{ppl}(y)$ term is down-weighted by 0.01 so that fluency remains secondary. We then keep the top-scoring output as chosen and the bottom-scoring output as rejected whenever the reward gap is at least 0.05.

Because reviewers reasonably questioned the perplexity term, we also ablate the coefficient in $\text{lex}(y) - w \log \text{ppl}(y)$ for $w \in \{0, 0.005, 0.01, 0.02\}$. This produces zero pair-order flips relative to $w = 0.01$ and changes the retained high-gap pool (≥ 0.4) only from 498 to 504 pairs, so perplexity acts only as a weak tie-breaker in the cleaned data rather than the main preference signal.

The legacy pipeline first selects the best SFT base among the mixed, UNK, and semantic variants using chrF++, then constructs preference pairs from that base. In the current snapshot that selector chooses the UNK model, yielding the legacy checkpoint `final_v2`, which serves as a negative control. The follow-up DPO runs instead start from the stronger multitask SFT model. After auditing the preference data, we keep only pairs whose reward gap exceeds either 0.2 (1,996 pairs) or 0.4 (498 pairs), and we train both standard sigmoid DPO and a robust+weighted variant [12, 13]. This turns the DPO family into a controlled test of base-model choice, pair filtering, and loss robustness.

4.4 Catalog-aware hybrid adjudication

The oracle analysis suggested that frontier and local candidates are complementary, but an oracle is not a deployable system. We therefore build a real inference-time adjudicator over the frontier DeepSeek baseline, UNK-SFT, and MT-DPO g0.4 sig. For each item it sees the Tangut source, dictionary glosses, the three candidate titles, their lengths, and simple prefix/suffix consensus hints.

The adjudicator uses Azure-hosted GPT-5.4 at temperature 0.0. Its prompt treats the task as historical catalog-title reconstruction rather than free paraphrase: dictionary glosses are clues rather than canonical modern titles, and short inputs are penalized for explanatory expansion. The model may either copy one candidate or perform a minimal repair using only candidate fragments or directly attested gloss segments. The repository stores the exact templates, frontier prompting script, fixed exemplar IDs, and per-example adjudication JSON, so the closed-model part of the workflow is at least auditable.

For reproducibility, we also add a deterministic open selector over the same 3-way pool. It never synthesizes text or calls an external model. It scores candidates using pairwise prefix/suffix consensus (shared prefix or suffix lengths counted only when they are at least two characters long), a title-suffix prior for final characters in {經論記疏頌儀義傳錄贊序字品文觀根次門}, deviation from the candidate-median length m , contamination penalties for ASCII spillover, angle brackets, [UNK], or known prompt artifacts, anti-truncation ($|y| \leq \max(1, |x|/4)$) and anti-expansion ($|y| \geq m + 5$ and $|y| \geq |x| + 5$) guards, and a small frontier prior. A guarded switch replaces the frontier output only when a local candidate has materially stronger conservative evidence. Table 3 later tests a 4-way extension that adds MT-SFT and shows that the original 3-way pool is already the best conservative operating point.

4.5 Training and judge details

All learned models start from Qwen2.5-7B-Instruct with LoRA adapters. SFT uses three epochs, learning rate 2×10^{-4} , batch size 4, gradient accumulation 4, rank-64 LoRA, and maximum sequence length 512. DPO uses two epochs, learning rate 5×10^{-5} , batch size 1, gradient accumulation 8, $\beta = 0.1$, rank-32 LoRA, cosine decay, and the same sequence length; the robust variants add label smoothing 0.1 and WPO-style weighting. The full pipeline fixes seed 42 for synthesis, data combination, and DPO training.

The reference-aware judge is a supplementary evaluator, not a training signal. It uses Azure-hosted GPT-5.4 to score reference agreement, source faithfulness, title-style fitness, and overall quality on a 1–5 scale after seeing the Tangut source, gold reference, and candidate simultaneously. Final scores are simple means over examples. For cross-judge robustness, the repository also includes a local Qwen2.5-7B-Instruct judge with the same rubric. We use the Azure judge as the primary table because it is the stricter and less saturated scorer.

5 Main Results

Table 1 separates three questions that earlier drafts blurred together: what the strongest prompt-only single model can do, what the strongest open/local trainable model can do, and what a real hybrid workflow can do. The first point is straightforward. The stronger frontier objection is real: the DeepSeek-V3.2 few-shot dictionary baseline remains the strongest single model.

The positive local-method result still lies inside the trainable regime. Structured multitask SFT is the best supervised base, and DPO helps only after it is moved onto that base and filtered much more aggressively. MT-DPO g0.4 sig is the strongest learned local model: raising the minimum reward gap from 0.2 to 0.4 shrinks the preference set from 1,996 to 498 pairs, lifts chrF++ from 20.93 to 30.01, and restores the length ratio from 1.74 to 1.12. The strict robust variant is slightly weaker on chrF++, and the legacy DPO checkpoint remains a genuine negative control.

The workflow result should be read separately from the single-model result. Table 3 now gives the consolidated workflow view. A fully open 3-way selector over frontier, UNK-SFT, and MT-DPO g0.4 sig already reaches 32.62 chrF++ and 2.90 reference-aware overall, so part of the gain is reproducible without closed adjudication. The 3-way catalog hybrid is the best conservative operating point at 34.87 chrF++, 3.02 overall, and 14/50 exact matches. Expanding the candidate pool with MT-SFT is informative but not better in the same sense: the 4-way catalog hybrid slightly raises chrF++ to 35.64, yet lowers overall to 2.98, exact matches to 13/50, and suffix preservation to 25/27. The 4-way open selector is also slightly worse than the 3-way open selector (32.47 vs. 32.62 chrF++; 2.70 vs. 2.90 overall). More candidates therefore raise overlap more easily than they improve conservative title reconstruction.

These disagreements are why the paper should not rely on a single automatic metric. chrF++, diagnostics, and the reference-aware judge reward different parts of the short-title problem. The defensible claim is therefore layered rather than absolutist: strict gap-filtered multitask DPO is the best local training recipe, and conservative 3-way adjudication turns that local complementarity into the strongest workflow.

The statistical analysis makes those layers more precise. Under 5,000 paired bootstrap re-samples, the best local DPO model improves over multitask SFT by +4.02 chrF++ with a 95% confidence interval of [0.84, 7.80]. More importantly for the workflow claim, the 3-way catalog hybrid improves over the frontier baseline by +6.33 chrF++ with a 95% confidence interval of [2.10, 11.67]. The gains in reference-aware overall (+0.22, 95% CI [0.00, 0.48]) and exact match (+0.04, 95% CI [0.00, 0.10]) are positive but boundary-level under this 50-example test set. The

Method	Lex $\uparrow$	PPL $\downarrow$	chrF++ $\uparrow$	Sem $\uparrow$	Flu $\uparrow$
<i>Prompt-only</i>					
Zero-shot	0.1165	20.21	0.19	1.00	1.56
Dict prompt	0.8187	4474.44	6.13	2.30	2.40
Dict+CoT	0.8723	18119.16	7.59	2.30	1.88
<i>Frontier DeepSeek</i>	0.6234	7438.72	28.54	2.62	2.78
<i>Local learned</i>					
Mixed-SFT	0.3192	298684.91	17.85	1.42	1.88
UNK-SFT	0.3514	11031.18	22.11	1.66	1.96
MT-SFT	0.4839	3322.93	25.99	1.70	2.26
Sem-SFT	0.3723	17064.38	17.03	1.46	1.84
Legacy DPO	0.4224	314577.16	19.39	1.52	1.76
MT-DPO g0.2 sig	0.5694	303438.00	20.93	1.88	1.86
MT-DPO g0.2 rob	0.5249	238892.42	27.67	1.88	1.88
MT-DPO g0.4 sig	0.4833	9573.71	30.01	1.84	2.04
MT-DPO g0.4 rob	0.4547	20012.69	27.52	1.74	2.08
<i>Hybrid workflow</i>					
<i>Catalog hybrid</i>	0.6189	6351.88	34.87	2.56	2.70
Human ref.	0.5733	1646139.34	100.00	2.24	2.34

Table 1: Repository-level metrics on the 50-example Tangut test set. Frontier DeepSeek is the strongest single-model baseline. The catalog hybrid adjudicates among the frontier output, UNK-SFT, and MT-DPO g0.4 sig. Here g0.2 and g0.4 denote minimum preference-gap thresholds. Bold marks the best local learned value in each column.

open selector is directionally better than the frontier baseline but not statistically robust at this sample size (+4.08 chrF++, 95% CI $[-2.28, 10.64]$). The right claim is therefore a robust overlap gain for the full hybrid, a reproducible directional gain for the open selector, and a conservative 3-way operating point rather than “more candidates is always better.”

6 Analysis

6.1 Reference-aligned diagnostics

Table 2 summarizes the output-side diagnostics. Here “contamination” flags outputs containing ASCII spillover, angle brackets, [UNK], or known prompt-artifact tokens, and “suffix preservation” is computed on references containing at least one canonical title-suffix character; a prediction is counted as correct when it preserves all such reference-side suffix characters. The table caption gives the exact suffix list.

The frontier DeepSeek baseline is already a clean single model: 12/50 exact matches, 0 contamination, 24/27 suffix preservation, and a near-reference length ratio of 0.95. Within the local family, multitask SFT is the cleanest pure SFT system, and strict gap-0.4 DPO regularizes the outputs further. The 3-way open selector stays clean as well (12/50 exact, 0 contamination, 23/27 suffix preservation, 0.98 length ratio), while the 3-way catalog hybrid is cleaner still with 14/50 exact matches, 26/27 suffix preservation, 0 contamination, and the same 0.95 length ratio as the frontier baseline. The workflow gain therefore does not come from verbose rewriting; it comes from more accurate conservative title reconstruction.

Method	Exact Match	Contam. Rate	Suffix Preserve	Len. Ratio
<i>Prompt-only</i>				
Dict prompt	0/50	0.00	3/27	1.83
<i>Frontier DeepSeek</i>	12/50	0.00	24/27	0.95
<i>Local learned</i>				
UNK-SFT	3/50	0.06	17/27	1.05
MT-SFT	5/50	0.02	19/27	1.10
Legacy DPO	2/50	0.18	18/27	1.88
MT-DPO g0.2 sig	5/50	0.00	19/27	1.74
MT-DPO g0.2 rob	4/50	0.14	23/27	1.58
MT-DPO g0.4 sig	5/50	0.04	22/27	1.12
MT-DPO g0.4 rob	5/50	0.04	22/27	1.26
<i>Hybrid workflow</i>				
Open selector (3-way)	12/50	0.00	23/27	0.98
<i>Catalog hybrid (3-way)</i>	14/50	0.00	26/27	0.95
Catalog hybrid (4-way)	13/50	0.00	25/27	0.95
Human ref.	50/50	0.00	27/27	1.00

Table 2: Reference-aligned diagnostics derived from stored predictions. “Contam.” denotes mixed-script or prompt-artifact contamination. “Suffix Preserve” is computed on references containing at least one title-suffix character in {經論記疏頌儀義傳錄贊序字品} and marks a prediction correct when it preserves all such reference-side suffix characters. Bold marks the best local learned value in each column, excluding prompt-only baselines, the hybrid workflow, and the human reference.

6.2 From complementarity to a deployable workflow

The oracle analysis mattered because it showed that frontier and local candidates were not near-duplicates. Table 3 replaces the earlier progression view with a more useful workflow comparison.

The 3-way deterministic selector over frontier/UNK-SFT/MT-DPO already changes 12/50 cases without any closed model or text synthesis: it selects the frontier 38 times, UNK-SFT 8 times, and MT-DPO 4 times. The 3-way catalog hybrid goes further. It selects the frontier 36 times, MT-DPO 6 times, UNK-SFT 5 times, keeps one tie, and performs only 2 minimal repairs. Thus 48/50 outputs come from direct candidate selection rather than free rewriting, and local models contribute directly on 11 cases plus two repairs. The exact-match union of the three candidate sources is also 14/50, so the hybrid closes the exact oracle gap completely even though the oracle still retains headroom on chrF++ and overall score.

Pool expansion is informative in the opposite direction. Adding MT-SFT as a fourth candidate slightly raises raw chrF++ under GPT adjudication (35.64) but lowers reference-aware overall (2.98), exact matches (13/50), and suffix preservation (25/27); the adjudicator then falls back to the frontier 41 times and uses MT-SFT only once. The 4-way open selector behaves similarly, dropping from 2.90 to 2.70 overall. Candidate diversity therefore saturates quickly: the original 3-way pool is not arbitrary, but the best conservative operating point in this snapshot. The qualitative cases make the same point. The workflow restores 五部經 instead of the frontier normalization 五類經, fixes the frontier truncation 番 to 番言金剛王乘根, and replaces the frontier paraphrase 德宜點義句顯宣論 with the more title-like 正理滴之句義顯了.

Workflow	Pool	Exact Match	chrF++	Overall
<i>Frontier DeepSeek</i>	single	12/50	28.54	2.80
3-way open selector	F+UNK+DPO	12/50	32.62	2.90
<i>3-way catalog hybrid</i>	F+UNK+DPO	14/50	34.87	3.02
4-way open selector	F+UNK+MT+DPO	12/50	32.47	2.70
4-way catalog hybrid	F+UNK+MT+DPO	13/50	35.64	2.98
Best-of-3 oracle	F+UNK+DPO	14/50	41.85	3.12

Table 3: Consolidated workflow comparison. “F” = frontier DeepSeek, “UNK” = UNK-SFT, “MT” = MT-SFT, and “DPO” = MT-DPO g0.4 sig. The deterministic selectors are fully open and never edit text. Adding MT-SFT as a fourth candidate slightly raises chrF++ under GPT adjudication but weakens exact match and reference-aware overall, so the 3-way pool remains the best conservative operating point.

6.3 Reference-aware judging changes the ranking

The legacy LLM judge in the repository is reference-free, so we reran evaluation with a stricter judge that sees the Tangut input, the gold reference, and the candidate simultaneously.

Table 4 changes the ranking. Among single models, the frontier baseline remains the clear winner at 2.80 overall. Within the local learned family, the best overall score is shared by three multitask-base DPO variants at 2.22. Among workflows, the open selector already reaches 2.90 overall and the 3-way catalog hybrid 3.02, while the 4-way hybrid slips back to 2.98. The gains are concentrated where catalog conservatism matters most: on non-suffix titles the frontier baseline scores 2.39 overall and the 3-way hybrid 2.74, and on references of length ≤ 6 the frontier scores 2.94 and the hybrid 3.24. A local Qwen2.5-7B judge preserves the same frontier/open/hybrid/human ordering (4.20, 4.20, 4.24, 4.90), reducing concern that the workflow ranking is purely a GPT-family artifact.

6.4 Metric alignment, robustness, and filtered DPO

The `human_reference` run is an important anchor. The gold reference receives perfect chrF++ and perfect reference-aware judge scores, but only moderate lexical coverage and catastrophic perplexity. Lexical coverage therefore measures overlap with dictionary-derived semantics rather than agreement with the gold title, while perplexity penalizes compact title style and historical transliterations. These metrics remain useful diagnostics, but they should not drive model selection in this task.

The bootstrap results in Table 5 clarify which claims are stable. The local chrF++ gain from multitask SFT to strict gap-0.4 sigmoid DPO remains above zero under resampling, and the catalog hybrid’s chrF++ gain over the frontier baseline does as well. The judge-overall and exact-match gains are directionally positive but boundary-level on this 50-example test set. The overlap audit addresses a separate concern: there is zero exact source, target, or full-pair overlap across train/dev/test; zero exact overlap between test references and the 50k synthetic multitask targets; and zero exact overlap between the frontier few-shot exemplars and the test references. This does not prove complete immunity from leakage, but it rules out the simplest exact-copy explanation.

The DPO audit explains why strict filtering helps. Across all 3,867 preference pairs, the reward-chosen candidate is closer to a simple gold proxy only 71.3% of the time, but this rises to 81.5% for gaps ≥ 0.40 . The high-gap filter is therefore isolating cleaner labels rather than acting as a cosmetic hyperparameter. The reward-weight ablation sharpens the same conclusion: varying the

Method	Ref. Agr.	Src. Faith.	Title Style	Overall
<i>Prompt-only</i>				
Dict prompt	1.50	1.52	2.10	1.50
<i>Frontier DeepSeek</i>	2.78	2.82	4.48	2.80
<i>Local learned</i>				
UNK-SFT	1.78	1.84	4.36	1.82
MT-SFT	1.92	1.98	4.18	1.98
Legacy DPO	1.64	1.72	3.80	1.68
MT-DPO g0.2 sig	2.08	2.24	4.14	2.22
MT-DPO g0.2 rob	2.10	2.16	3.78	2.12
MT-DPO g0.4 sig	2.12	2.26	4.60	2.22
MT-DPO g0.4 rob	2.10	2.20	4.70	2.22
<i>Hybrid workflow</i>				
Open selector (3-way)	2.82	2.88	4.50	2.90
<i>Catalog hybrid (3-way)</i>	2.94	3.00	4.72	3.02
Catalog hybrid (4-way)	2.90	3.00	4.64	2.98
Human ref.	5.00	5.00	5.00	5.00

Table 4: Reference-aware judge scores on a 1–5 scale. The judge sees the Tangut input, the gold reference, and the candidate output simultaneously. Bold marks the best local learned value in each column, excluding prompt-only baselines, the hybrid workflow, and the human reference.

Comparison ($B - A$)	Δ chrF++	Δ Overall	Δ Exact
3-way catalog hybrid – Frontier	+6.33 [2.10, 11.67]	+0.22 [0.00, 0.48]	+0.04 [0.00, 0.10]
3-way open selector – Frontier	+4.08 [−2.28, 10.64]	+0.10 [−0.20, 0.40]	0.00 [−0.08, 0.08]
Local – MT-SFT	+4.02 [0.84, 7.80]	+0.24 [0.00, 0.46]	0.00 [0.00, 0.00]
Local – Frontier	+1.47 [−8.09, 10.29]	−0.58 [−1.04, −0.12]	−0.14 [−0.26, −0.04]
Legacy DPO – MT-SFT	−6.60 [−11.84, −1.73]	−0.30 [−0.64, 0.04]	−0.06 [−0.14, 0.00]

Table 5: Paired bootstrap comparisons with 5,000 resamples. “Overall” denotes mean reference-aware judge score, and “Exact” denotes exact-match rate.

coefficient on log ppl from 0 to 0.02 causes zero pair-order flips relative to the current reward and changes the gap-0.4 pool only from 498 to 504 pairs. In other words, the small perplexity term is not selecting different high-gap pairs; the real stabilizer is still aggressive gap filtering on top of the multitask base. Once DPO is moved onto that base and filtered at 0.4, it stops behaving like a blanket negative result and becomes the strongest local model. The hybrid workflow does not erase this point; it depends on the local candidate pool being strong enough to contribute conservative alternatives in the first place.

7 Discussion and Limitations

The practical reading of the paper is simple. If a user can make only one model call, the DeepSeek-V3.2 frontier baseline is the best single system we measured. If the target use case is scholar-facing short-title reconstruction, however, the stronger claim is the workflow claim: conservative local candidates help because they sometimes preserve short historical title chains that frontier prompting normalizes, truncates, or over-regularizes.

That distinction makes the contribution more general, not less. The local models matter not

Audit item	Value
Total preference pairs	3867
Duplicate chosen/rejected pairs	3
Non-finite reward entries	3
Mean reward gap	0.2396
Chosen closer to gold proxy	71.3%
Chosen closer on gap (0.05, 0.10]	62.3%
Chosen closer on gap (0.20, 0.40]	75.4%
Chosen closer on gap ≥ 0.40	81.5%
Pairs kept with gap ≥ 0.20	1996
Pairs kept with gap ≥ 0.30	1010
Pairs kept with gap ≥ 0.40	498

Table 6: Lightweight audit of the current DPO pair construction pipeline. Higher reward gaps correlate with cleaner preference labels under a simple gold-proxy check, which helps explain why strict filtering stabilizes the later multitask-base DPO runs.

because they beat a frontier model head-to-head, but because they diversify the candidate pool with structurally conservative hypotheses. A similar recipe should transfer beyond Tangut book titles: train local models to stay close to title structure, then adjudicate with a title-aware reranker rather than a generic fluency prior.

The scope should also be explicit. The current dataset supports Tangut short-text translation, especially title-like strings, not general open-domain Tangut sentence translation. The synthetic corpus is a structural training proxy rather than realistic pseudo-Tangut, and the strongest gains are concentrated on short and mid-length items. We therefore treat multilingual-transfer and frontier-only sampled reranking baselines as important next controls rather than settled negatives.

Two limitations matter most. First, the real dataset is extremely small, so judge and exact-match gains remain borderline even when chrF++ gains are robust. Second, the primary adjudicator and judge are closed models, and we do not yet provide blinded expert validation. The deterministic open selector narrows this reproducibility gap but does not close it; the repository should therefore release the frontier prompt script, exemplar IDs, adjudication JSON, and per-item predictions alongside the paper. The most useful next steps are a sampled frontier-only MBR/QE reranking control, distilling the adjudicator into an open local reranker, validating the judge on a small expert-annotated subset, and testing transfer to other historical short-form datasets.

8 Conclusion

The final message of this paper is now more precise and more useful. Strong frontier prompting does not fail on Tangut short-text translation; it is the strongest single-model baseline. Within the open/local regime, however, structured multitask supervision remains the right base recipe, and aggressively filtered multitask-base DPO produces the strongest local model. More importantly, those local models are not merely weaker copies of the frontier output. When their complementary historical-title hypotheses are combined through catalog-aware adjudication, the resulting workflow becomes the strongest non-reference system overall, reaching 34.87 chrF++, 3.02 reference-aware overall, and 14/50 exact matches.

This changes the practical lesson. For ultra-low-resource historical short-text translation, the right question is not simply whether local models can replace a frontier API. It is whether structured local training can produce conservative alternative hypotheses that improve a scholar-facing

workflow. In this repository, the answer is yes. The broader methodological lesson is equally important: evaluation must be anchored to reference-aware criteria, because generic reference-free metrics can badly mis-rank compact historical titles and even penalize the gold reference itself.

References

- [1] Thea Sommerschild et al. 2023. Machine Learning for Ancient Languages: A Survey. *Computational Linguistics*, 49(3).
- [2] Wang et al. 2023. Findings of the EvaHan 2023 Workshop on Ancient Chinese Translation. In *Proceedings of the EvaHan 2023 Workshop on Ancient Chinese Translation*.
- [3] Miguel Domingo, Mercedes Garces-Diaz, Francisco Casacuberta, and Rafaela Perez-Ortiz. 2018. A Machine Translation Approach for Modernizing Historical Documents. In *Proceedings of the Language Technologies for All Workshop*.
- [4] Rico Sennrich, Barry Haddow, and Alexandra Birch. 2016. Improving Neural Machine Translation Models with Monolingual Data. In *Proceedings of ACL*.
- [5] Chen et al. 2017. Incorporating Bilingual Lexicon into Neural Machine Translation. In *Proceedings of ACL*.
- [6] Jiatao Gu et al. 2018. Universal Neural Machine Translation for Extremely Low Resource Languages. In *Proceedings of NAACL*.
- [7] Chris Hokamp and Qun Liu. 2017. Lexically Constrained Decoding for Sequence Generation Using Grid Beam Search. In *Proceedings of ACL*.
- [8] Surangika Ranathunga et al. 2021. Neural Machine Translation for Low-Resource Languages: A Survey. *ACM Computing Surveys*, 54(11).
- [9] Hongyuan Lu et al. 2023. Chain-of-Dictionary Prompting Elicits Translation Robustness in Large Language Models. *arXiv preprint arXiv:2305.06575*.
- [10] You-Cheng Liao et al. 2024. Learning-From-Mistakes Prompting for Indigenous Language Translation. In *Proceedings of the Seventh Workshop on Technologies for Machine Translation of Low-Resource Languages*.
- [11] Rafael Rafailov et al. 2023. Direct Preference Optimization: Your Language Model is Secretly a Reward Model. *arXiv preprint arXiv:2305.18290*.
- [12] Jin Chen et al. 2024. Provably Robust DPO: Aligning Language Models with Noisy Feedback. *arXiv preprint arXiv:2403.00409*.
- [13] Weiran Pang et al. 2024. WPO: Enhancing RLHF with Weighted Preference Optimization. *arXiv preprint arXiv:2406.11827*.
- [14] Yang et al. 2024. DQO: Direct Preference Optimization with Query-specific Rewards for Neural Machine Translation. *arXiv preprint*.
- [15] Bogoychev et al. 2025. Improving Neural Machine Translation with Preference Optimization via Word Alignment and MBR Decoding. *arXiv preprint*.
- [16] Maja Popovic. 2017. chrF++: Words Helping Character n-grams. In *Proceedings of the Second Conference on Machine Translation*.
- [17] Ricardo Rei et al. 2020. COMET: A Neural Framework for MT Evaluation. In *Proceedings of EMNLP*.

- [18] Vilem Zouhar et al. 2024. Pitfalls and Outlooks in Using COMET. In Proceedings of WMT.
- [19] Jose Pombal et al. 2025. Adding Chocolate to Mint: Mitigating Metric Interference in Machine Translation. arXiv preprint arXiv:2503.08327.
- [20] Fabio Kepler et al. 2019. OpenKiwi: An Open Source Framework for Translation Quality Estimation. In Proceedings of ACL.